Approximate Radix-4 8-bit Booth Multiplier for Efficient DNN Acceleration

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Abstract: Deep Neural Networks (DNNs) rely heavily on multiply accumulated operations, making multiplier efficiency critical for reducing processing cost. This work evaluates an approximate radix 4 Booth multiplier where the five least significant partial product columns are replaced with a fixed pattern (10000). This simplification reduces hardware area and switching activity while keeping the most significant bits accurate for inference. Results show meaningful reductions in logic complexity and power compared to exact Booth designs, with a small loss in accuracy, offering an effective tradeoff for resource constrained accelerators.

Keywords- Approximate computing, Booth multiplier, deep neural networks, hardware acceleration, radix 4 multiplication

I. Introduction

Deep Neural Networks (DNNs) deliver high accuracy but require a large number of multiply and accumulate operations, making them expensive in terms of power, delay, and hardware area. Since many DNN workloads can tolerate small numerical errors, approximate computing has become an effective way to reduce the cost of arithmetic units. This review focuses on an approximate radix 4 Booth multiplier that replaces the five least significant partial product columns with a fixed pattern to improve efficiency while keeping acceptable accuracy for inference.

II. Applied Approximate Technique

This project implements an approximate radix 4 Booth multiplier by modifying the five least significant partial product (PP) columns. In a standard radix 4 Booth design, each 3-bit multiplier group generates one of {0, ±M, ±2M}, which is aligned and accumulated to form the 16-bit product. While accurate, the lower PP columns contain dense overlaps that increase switching activity and power, and they contribute to adder depth that affects the overall critical path.

To reduce this cost, the five LSB PP columns are replaced with a fixed pattern (10000), eliminating register widths and adder hardware in this region. Only the upper columns (bits 5 to 15) use exact radix 4 Booth encoding. Since these higher significance bits dominate the product's magnitude, the approximation is sufficiently reasonable and reliable in reduced logic complexity and power with only a small loss of accuracy in inference applications

Table 1 shows the fixed pattern of substitution used in the approximate design. Columns 0 to 4 are replaced with the constant pattern 10000, while Columns 5 to 15 follow exact radix 4 Booth rules. This selective truncation eliminates some hardware in the LSB region, reducing switching activity and shrinking partial product accumulation.

Column Index	Exact Booth PP Output	Description
0	Fixed pattern 10000	LSB replaced with constant (1)
1	Fixed pattern 10000	Fixed pattern 10000
2	Fixed pattern 10000	Fixed pattern 10000
3	Fixed pattern 10000	Fixed pattern 10000
4	Fixed pattern 10000	Fixed pattern 10000
5–15	Booth-encoded PP bits	Higher-order bits remain exact

Exact vs Approximate Partial Product Generation

Table 2 lists the standard radix 4 Booth encoding applied to Columns 5 to 15. These mappings (0, ±M, ±2M) remain unchanged in the approximate design, ensuring that all high significance bits are computed exactly. Preserving correctness in the upper columns prevents large numerical deviations and allows the truncated multiplier to sustain near exact inference accuracy for MAC dominated DNN workloads, while still achieving meaningful hardware savings.

d(i+1) d(i) d(i−1)	Partial Product
000	0
001	+1 x Multiplicand
010	+1 x Multiplicand
011	+2 x Multiplicand
100	-2 x Multiplicand
101	-1 x Multiplicand
110	-1 x Multiplicand
111	0

Standard Radix-4 Booth Encoding

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III. ACCURACY ANALYSIS

The approximate radix 4 Booth multiplier was evaluated against the exact radix 4 multiplier using 3×3 convolution operations on 8-bit quantized feature maps. The exact radix 4 design achieves 96 percent inference accuracy and serves as the reference baseline. The proposed approximate LSB-10000 design introduces controlled truncation by replacing the five least significant partial product columns with a fixed pattern, affecting only the LSB region of the product.

Experimental results show that the truncated design maintains 95 percent accuracy on the last convolution layer, and 84 percent overall accuracy across all layers based on our sample neural network. Since truncation does not alter the MSBs, numerical deviations remain bounded in magnitude, producing an NMED value of 0.02872 hence still maintaining acceptable inference performance until the last layer.

In terms of hardware efficiency, the approximate design reduces area by about 12 percent and power by about 11 percent relative to the exact radix 4 multiplier, due to the removal of Booth encoders, sign extension logic, and adders in the LSB region. Although the critical path increases slightly (0.81 ns to 0.87 ns) due to synthesis tool component selection, the tradeoff is favorable for MAC dominated inference where parallelism is often preferred for neural network workloads.

IV. HARDWARE IMPLEMENTATION

The design uses a modular 8-bit approximate radix 4 Booth multiplier with a top-level RTL module that takes two signed 8-bit inputs and outputs a 16-bit product. Columns 5 to 15 follow the standard radix 4 Booth flow: the multiplier bits are grouped in threes, and the Booth encoder generates the correct multiples of the multiplicand (0, ±M, ±2M).

In the approximate region (columns 0 to 4), we discard as much hardware as possible in those regions, namely in the partial product generation. These five LSB partial product columns are replaced with the fixed pattern 10000, eliminating logic and reducing switching activity. The remaining exact partial products are summed with a simplified adder tree to form the final output.

This structure preserves accuracy in the high significance bits while reducing hardware complexity and energy cost. Although the critical path increases slightly due to side effects from truncation, the savings in area and power make the design well suited for inference tolerated DNN accelerators.

V. RESULTS/ SIMULATION RESULTS

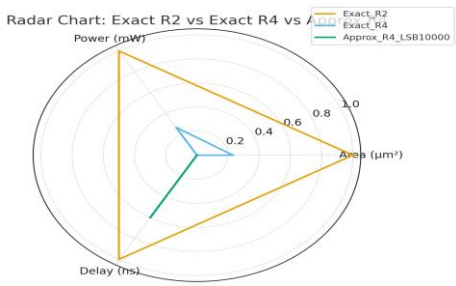
Metric	Exact_R2	Exact_R4	Approx_R4_LSB10000
Area (μm^2)	751.7	537.5	473.0
Power (mW)	0.1081	0.08252	0.07325
Delay (ns)	0.91	0.81	0.87
All Layers Accuracy (%)	96	96	84
Last Layer Accuracy (%)	96	96	95
NMED	0	0	0.02872

Table 1. Post-synthesis and inference results for exact and approximate radix-4 Booth multipliers.

Table 1 compares three multiplier designs using post synthesis metrics and DNN inference on 8-bit 3×3 convolutions. The exact radix 4 Booth multiplier achieves 96 percent inference accuracy and serves as the reference baseline. An exact radix 2 Booth implementation shows higher hardware cost, while providing the same accuracy, due to additional partial product generation.

The proposed approximate design (Approx_R4_LSB10000) achieves the lowest hardware cost by replacing columns 0 to 4 with a fixed pattern. Relative to the exact radix 4 multiplier, it reduces area by about 12 percent (537.5 µm² to 473.0 µm²) and reduces power by about 11 percent (0.08252 mW to 0.07325 mW). The delay increases slightly from 0.81 ns to 0.87 ns due to the compiler choosing different components in the final stage.

Accuracy remains suitable for inference tolerant workloads: the last layer classification accuracy stays at 95 percent, and overall pipeline accuracy remains 84 percent. Truncation produces a non zero error (NMED = 0.02872), but the error remains bounded within the LSB region, preserving meaningful DNN behavior.



Comparative Performance Evaluation of Radix-2, Exact Radix-4, and Approximate Radix-4 Booth Multipliers

CONCLUSION

This work evaluated an approximate 8 bit radix 4 Booth multiplier that replaces the five least significant partial product columns with a fixed pattern to reduce hardware cost. By keeping exact Booth encoding only in the higher order bits, the design lowers area and power while maintaining useful accuracy for DNN inference. Although the delay increases slightly, the efficiency gains outweigh the overhead for inference tolerant workloads in parallelism. These results show that selective LSB truncation can provide meaningful hardware savings with minimal impact on output quality, making it suitable for low power MAC based accelerators.

REFERENCES

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